KATULA VAISHNAVI

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SUMMARY

B.Tech Data Science student (2027) with hands-on experience in Artificial Intelligence, Machine Learning, Data Science, Data Analysis, Predictive Modeling, and Data Engineering. Experienced in data gathering, data exploration, data transformation, ETL pipelines, model evaluation, statistical analysis, and preparing analytics-ready datasets using Python, SQL, Scikit-learn, TensorFlow, and cloud technologies. Passionate about solving business problems through data-driven decision-making and scalable AI solutions.

EDUCATION

Malla Reddy Deemed to be University — Hyderabad, Telangana
B.Tech in Data Science (CGPA: 7.81) — *Aug 2023 – Jun 2027*

Intermediate (Class XII) — *Percentage: 83.7%*
Secondary School (Class X) — *Percentage: 86.04%*

EXPERIENCE

Edunet Foundation — Hyderabad
AI and Cloud Intern — *Jun 2025 – Aug 2025*

- Built multi-model Disease Prediction system benchmarking Logistic Regression, Random Forest & XGBoost.
- Performed feature engineering, handled missing/imbalanced data & hyperparameter tuning for better recall.
- Evaluated models with precision, recall, F1, ROC-AUC and presented SHAP-based clinical explainability.
- Deployed the model via Flask API and documented the pipeline using AWS cloud fundamentals.

TECHNICAL SKILLS

Languages: Python, SQL
ML & AI: Scikit-learn, PyTorch, TensorFlow, RAG, Prompt Engineering, Generative AI (LLMs), Predictive Modeling
Data, DB & Tools: Pandas, NumPy, ChromaDB, LangChain, NoSQL, ETL, Power BI, Matplotlib, Seaborn, Statistical Analysis, Exploratory Data Analysis, Data Mining
Engineering: FastAPI, Flask, Cloud (Basics), Streamlit, React, Docker, Git, AWS, Azure

PROJECTS

AmazonQA-RAG | Python, QLoRA, FAISS, LangChain, RAGAS — GitHub

- Fine-tuned LLM via QLoRA (4-bit NF4 + LoRA) for low-resource instruction adaptation on Amazon Q&A data.
- Built FAISS-backed RAG pipeline with sentence-transformers; benchmarked Base/RAG/Finetuned on RAGAS.
- Optimized retrieval latency & context handling, achieving 28% better answer relevancy over base RAG.
- Added streaming response support and citation highlighting in the Streamlit interface for better UX.

Retail Sales Data Warehouse | Python, PostgreSQL, Airflow, dbt, Docker, Power BI — GitHub

- Built star-schema DWH (SCD Type 2) processing 10K rows / $2.3M revenue with 10 data quality checks.
- Delivered 3 business marts (RFM, inventory turnover, daily KPIs), cutting reporting time from 4 hrs to 8 min.
- Automated incremental loads & dbt tests with Dockerized Airflow, reducing manual work by 90%.
- Created interactive Power BI dashboards with row-level security for different business stakeholders.

Matrix Factorisation Recommender | Python, PyTorch, SVD, NCF, BPR — GitHub

- Implemented & benchmarked 5 models (SVD, NCF, BPR vs Popularity, Item-kNN) on NDCG@10 / MAP@10.
- Designed cold-start evaluation and ran ablations on embedding size, negative sampling & dropout via FastAPI.
- Achieved 15% NDCG@10 lift over strongest baseline and packaged inference service with Docker.
- Added real-time recommendation endpoint with caching and A/B testing support for model comparison.

CERTIFICATIONS

- Google GenAI AI/ML – Hack2Skill, 2025
- Introduction to Technology and AWS – Coursera, 2025
- Microsoft Azure AI-900 – Certiport / PearsonVUE, 2023
- Infosys Pragati Cohort – SpringBoard, 2025
- AWS Learning: Generative AI – AWS Skill Builder, 2026